Chapter 13 : Anatomy for Skateboarding III

Introduction

The third and final chapter on anatomy for skateboarding outlines specific exercises used to remedy persistent soreness. The first muscular weakness was right lower back pain from the participant's view and occurred prior to skateboarding in high school while performing heavy back squats. Right calf pain also occurred during this time after extensive running and walking. The next muscular pain developed in early 2022 near the right anterior lateral knee from running after several years of low exercise. Near the end of the second year skateboarding, right lateral knee pain was mostly eliminated and replaced with lower calf pain. Shortening and isometrically holding the gastrocnemius followed by lengthening instigated foot muscle soreness that was similarly alleviated. After recorded pain data from the previous two chapters, synopsis of exercising is included from December 2024 until February 2026 to articulate the resolution for persistent lower lateral quadricep and soleus soreness.

Methods

There were several muscular issues that decreased enjoyment in athletics. Right posterior lower leg pain appears while dehydrated during unusually long exercises in variable barometric pressure and was resolved from shortening the gastrocnemius. Greater right to left anterior and posterior thigh muscles were from learning heavy back squats and emerged learning switch ollies with excessive left leg raise over the tail. Right lower back pain was resolved with regular stanced distance longboarding improving hip stability through increased left gluteus medius strength. Gluteus strength decreases while sitting most of the day at a sedentary office job and led to dead butt syndrome (DBS), iliotibial (IT) band snap, and right lower anterior quadricep pain. Head and neck pain is easier to guess since it is related to diet and hydration. Posture related muscle differences were addressed by lifting and using the other limbs.

Lower leg

Medical history includes a partially torn talofibular ligament and it was uncertain if the lateral ankle sprain in 2010 had lasting issues since there was audible noise anteriorly while rotating the foot. Posterior tendon popping during ankle rotation began with skateboarding tricks near the lower medial section of the calcaneal tendon. Tom, Dick, and Harry and in particular the flexor hallucis longus are implicated tendons from overuse inflammation causing audible posterior ankle noises. Kicking the rugby ball with the right ankle dorsiflexion muscles caused anterior inflammation but the same medial popping was not observed on the left ankle dorsiflexion muscles while kicking with the left leg. Audible noise continued after ball kicking

cessation and was ruled out as the cause. After the 2024 data collection period ended for the pain charts, effort was extended in 2025 to use the left leg with switch skateboarding, longboarding, and ball kicking. These activities resulted in identical anterior left ankle sound while the anterior right ankle ceased. Cross leg sitting with the right ankle underneath the left knee causes ankle inversion and results in similar posterior ankle popping when feet are reversed. While sitting in a chair, the left ankle unconsciously moves on top of the right knee, placing weight on the left buttocks.[2]

The right posterior lower limb experienced cramps and was sore following school organized American football from 2008-2011. The same right soleus muscle soreness occurred after back squats in the two weight lifting periods in 2024 and 2025. This effect might be gait related since walking and running was consistently performed as inflammation reduction (*Chapter 12 - Figure 9*) for the hamstrings. Walking and running in partially melted snow or ice illuminated exaggerated lateral hip movement imbalance. Right medial posterior lower leg is the failing point while longboarding during the last half of lengthy regular stanced heelslides and the front left leg is forced to step backwards off the board. Failing switch heelslides displayed a pronounced weight falling forward by standing on the toes and identical overcorrecting by stepping off and backwards with the right front leg. An equivalent muscle observation occurred during switch FS 180s when raising the stronger rear anterior left leg off the board, resulting in unintentionally quarter flipping the board counterclockwise on landing. Failing to move the right front leg fast enough from switch pop shove its rotated the board clockwise and is evidence of weaker or less coordinated right quadriceps. The muscles used in this movement are the sartorius to rotate the thigh laterally and pectineus to adduct.[1]

One specific exercise that stretches the right anterior lateral lower leg to counteract the pronated ankle and subsequent knee valgus jump during regular stanced ollies included regular stanced Casper stalls. Another similar exercise leveraged during this period was rolling the ankle supine and forward into the board during moving skateboard manuals as suggested by internet video blogs to learn the kickflip. These movements are similar to modern physical therapy techniques such as squatting or jumping from a bosu ball. The end of May in 2025 included a skateboarding break where 20 miles of running were logged in five days and a loud synovial fluid or inflamed tendon over bone noise occurred in the left lateral dorsal peroneal ankle muscle suggesting more attention should be paid to ankle flexibility. Switch downhill longboarding practice began in ernest progressing from small carves at the end of May 2025 to longer standing heel and toeslides with the intention of improving both ankle dorsiflexion and left adductor to right abductor symmetry. Medial lower leg soreness on both legs was observed after an extended period in 2025 and switch heelslides are not consistently smooth, returning parallel downhill throughout the last half of the heelslide in a similar manner from lack of experience. The end of November in 2025 also noted a sore left soleus from standing and that regular stanced tailslides increased right hallucis extensor soreness.

After outdoor skateboarding ceased in November 2025, right posterior lower leg soreness sporadically reappeared despite normalizing muscle imbalances with weight lifting. Regular calf maneuvers seemed to have little effect and walking to warm up before stretching further increased inflammation. Shortening the upper gastrocnemius and muscles behind the patellar such as the popliteus followed by lengthening those muscles relieved the persistent pain. Less intense methods that were part of the daily flexibility routine were ineffective and relief required laying on the side of a bed and isometrically flexing the sore leg in hip extension and knee

extension with a band. Full alleviation of the resulting lower soleus and foot pain occurred in February 2026 with a reduction in exercise intensity and dragging the toes across the ground into a shin high kick identical to the ollie flips. Instigating knee pain from excessive exercise is not required since either leg is now able to kickflip.

Quadriceps, hamstrings, and gluteus, back squats

High weight and low repetition backsquats in late 2024 displayed a slight weight shift from center to right during leg flexion and right to left in extension while exhausted and is partially caused by weaker left abductor and right adductors. Squat technique was improved through rolling the hips and gluteus posteriorly to limit knee snapping. Strategies to improve the ollie revealed an increase in rear leg posterior adductor strength during knee flexion in the descent and anterior abductor strength during knee extension or jumping. When the lateral right thigh muscles are inflamed, the inflexible IT band framework snaps over the lateral epicondyle of the femur. Similar snapping was eventually observed with the vastus intermedius and rectus femoris around the patellar from a combination of exhaustion and seasonal temperature variation. The left quadricep snaps near the knee while lengthening during the eccentric descent portion of the squat and the right quadricep snaps during the concentric ascent portion when the muscle shortens indicating stronger left anterior muscles compared with the right leg.

Right lower back pain occurred during heavy weighted and forward leaning powerlifting back squats in school that relied on the stronger right gluteal muscles to complete the lift. Since there were several medical visits, lower back pain was more of an event and forward lean technique could have been unconsciously avoided in the decade of intermittent lifting in the

interim. When resuming weightlifting during the 2025-2026 winter, the internet suggested improving back squat technique by using a narrower grip and pulling the bar into the trapezius with more forward chest lean and posterior flexed shoulders. Increased strength and flexibility along with improved squat form eliminated knee pain and decreased audible noise. The inclusion of cardiovascular conditioning initiated the return of knee snap and supports gait related deformity while running from hyperextending the right knee.

The right leg also displayed chaffing after incline hill running compared with left leg chafing when the right leg was consciously lifted higher on the same route on a later date. Sole concentration on limiting hyperextension on the right knee leads to the aforementioned left iliopsoas snap from Chapter 11. Muscle imbalance theory partially caused by squatting is supported by performing back squats with slight weight shift onto the left leg and forward to use the gluteal and hamstring muscles, resulting in identical left lower back muscular soreness. This information could also be used to support the idea for stronger left quad to right quad or less left leg flexibility than right leg flexibility because the right posterior muscles are used more during exercise. A shortened posterior gluteus and hamstring group rotates the pelvis backwards, increasing quadricep flexibility while recovering but producing noticeable snaps.

Quadriceps, hamstrings and gluteus, longboarding and skateboarding

Physical therapy recommended lateral leg raise exercises in 2009. Regular stanced longboard pushing from 2014-2016 eliminated right lower back soreness but forced more pressure onto the vastus lateralis while resuming exercise in 2020. A similar weight lifting exercise for the gluteus medius is single leg step ups or side wall sits as prescribed from physical

therapy. The lateral knee attachment destabilizes the patellar when the vastus lateralis and gluteus medius are inflamed from overuse. While learning to push the longboard for transportation in 2014, the left gluteus medius muscle size increased and reduced lateral hip movement and right lower back pain. The right gluteus was also compressed from the right push leg. Although the primary weight lifting exercise for legs changed from back squats in 2006-2011 to light weight front squats in 2014-2016, significant effort was spent on olympic style squat clean from 2011-2013 and resulted in right lower back pain indicating that longboarding was the difference. Carves and slides downhill using a longboard builds upper adductor and gluteus strength that is necessary to balance the overdeveloped lower and lateral thigh muscles from ollies. Walking uphill builds gluteous and upper hamstring muscle fibers in a direction that is avoided by skateboarding alone and results in less soreness compared with squat related weight lifting.

IT band snap and lateral knee pain is related to underused gluteal muscles and is referred unofficially on the internet as Dead Butt Syndrome (DBS). Posterior rotated pelvis sometimes occurs while sitting for long periods during a desk job and reduces gluteus medius strength. Gluteus maximus strength also decreased significantly which reduced lower back muscle tension but allowed the pelvis to rotate unconsciously causing lower lateral quadricep pain and skateboard ollies overworked these muscles. Learning the ollie in early 2024 resulted in lateral quadricep and gluteous inflammation with reduction attempts by stretching the IT band through cross leg bends pulling the popliteus.

Regular stanced backside 180 ollies were difficult because the skateboard is landing switch and blind to the front right foot on landing. The excessive jumping attempts in August 2024 during this trick led to an overdeveloped lower bicep femoral muscle. Lower limb stability

from downhill longboarding is helpful for rotational ollies. Lower body pain reduction was the primary focus of longboarding since skateboarding anteriorly rotates the pelvis, lengthening the quadriceps and reducing hamstring flexibility. Shortened anterior or posterior lower leg muscles would pull the patellar bone forward or backwards and lengthen thigh muscles. The November-February 2024-2025 back squat period noted increased popliteal snap over the posterior lateral condyle of the femur similar to the lateral quadricep noise.

The primary form of exercise and transportation from 2017-2021 was walking and muscle imbalance contributed to right knee pain since the femur rotates medially and hyperextends the leg. Sedentary desk job sitting relaxes the gluteous muscles leading to lower back pain or knee pain and sitting with cross legs can lead to muscular asymmetry.[2] Prescribed medication during this period caused immobilization during sleep and side resting by unconsciously facing away from heat or cooling air flow would exacerbate the already overworked right leg from poor stride mechanics. While viewing historical and current televised sports to observe knee valgus while turning and landing, it was observed the players sat with knees and feet open though it might be from chair proximity. Anecdotally, amateur athletes in school are witnessed sitting with a variety of orientations while not being recorded.

Extensive regular stance distance longboarding from 2014-2016 increased right bicep femoris strength and attempting heelslides developed the posterior or magnus adductor. Longboard slides increase adductor strength for BS or FS 180 skateboarding ollies by reasoning of heelslides pushing the board perpendicular to the road with hip extension and toeslides pulling the board against the road with hip flexion. After extensive longboarding near the end of 2025, the thighs were more muscularly balanced and the external hip rotators were more involved in the 180 rotations. The left posterior was sore after landing regular stance tricks and partially

explains the left rear leg lifting off the tail when performing switch stance tricks. It was noted that standing for a significant part of the day accumulated soreness in the right in 2024 that lower back and hip muscles instead of the right lateral lower vastus lateralis. The concept is important since stretching the hamstrings with standing bends by posteriorly rolling shoulders produces lower back pain instead of bowing the head that strains the lateralis and trapezius resulting in neck or head pain.

Head and neck

Head and neck pain decreases enjoyment and frequency to skateboard and the goal is to limit unnecessary straining. One possible explanation for higher head pain prediction accuracy (*Chapter 12 - Figure 20*) compared with the other body parts is more thorough knowledge about what causes head pain compared with lack of experience exercising the lateral quadriceps through skateboarding (*Chapter 12 - Figure 24*). Head and neck pain originated from stress and dehydration (*Chapter 12 - Figure 26* and *Figure 27*) and is well documented in the participant's medical history. The origins during skateboarding were from jumping and landing with the rhomboids rolled forward into the latissimus dorsi while looking at the board but there are several other sports that primarily have the neck and head down. Traditional olympic and powerlifting techniques to strengthen upper back and chest muscles helped to correct posture during the two winter lifting sessions, however head pain emerged from excessive forward lean while exercising the posterior muscles. Handstand pushups in particular strain the neck and trapezius while looking at the ground instead of retaining a neutral position identical to the overhead barbell press. Nutrition is essential to avoid unnecessary head pain and proves

challenging since extensive muscle rebuilding molecules and water accumulate in the gluteus medius, obliques, and abdominis rectus rendering scale weighing or superficial cues for health subjective.

Unnecessary bodily ailments appeared from several related causes. Unconsciously using one limb for specific tasks resulted in a sore right calf that was rectified by shortening the gastrocnemius. Lower back pain was a problem while learning back squats and was reduced from longboarding. Gluteus strength developed while in school decreased while taking antipsychotics at a sedentary office job and rapidly exposed lower knee pain. Head and neck pain limits enjoyment while exercising. Inadequate hydration and nutrition are difficult while tired from exercising or in a caloric deficit from losing weight and finding the correct exercise to reduce soreness is challenging.

Results

There were three distinct exercise regiments repeated twice from the start of learning skateboarding ollies in January 2024 until January 2026. The intention was to learn how to kickflip consistently and proved challenging while also increasing muscular balance through improved athleticism. Distance and downhill longboarding ability took three summers of five days a week riding to attain prior to skateboarding and should not be dismissed as unnecessary. The first phase is titled *Outdoor Skateboarding* and was repeated twice. Practice commenced from January 2024 until November 2024 with increased pacing from July 2025 until December 2025. The next phase is called *Lifting and Plyometrics* and was practiced with a limited 275 pound freeweight set from December 2024 until February 2025 and from December 2025 until February 2026 to regain lower limb muscular balance and reduce right lower quadricep pain. The final exercise cycle is called *Indoor Skateboarding* and was practiced with a variety of boards in the occupants basement from February 2025 until July 2025 and from February 2026 until May 2026.

Outdoor Skateboarding

The frequency of outdoor skateboarding practice increased with improvements in mental stamina and muscular endurance. The quality of skateboarding also scaled during this time where the first few months in 2024 involved 30-45 minutes of low and slow rocket ollies but the end in 2025 saw improvements in both stanced ollies, FS 180, BS 180, pop shuvit, front shuvit, and erratic kickflips and heelflips. The first year had a maximum of three skateboarding practices a

week for six weeks until lower right knee pain forced two to three week breaks. The second year included learning switch downhill longboarding slides and improving regular stanced standing toeslides. August until November saw the addition of almost daily downhill longboarding, skateboarding, and evening hill walking before stretching while learning the kickflip.

Lifting and Plyometrics

The second phase includes weight lifting and plyometrics justified by cold temperatures persisting from November until the end of March. The climate in the mid-latitude continental United States also typically has freezing precipitation or snow. The perception of winter is unenjoyable and there would be a five month break until practice resumed. Common ailments include readily quantifiable eczema and low vitamin D and stem from a low awareness indoor lifestyle featuring dry heated air and low sunlight. Another complaint that is less easily monitored is psychological well being from low pollen. There are not that many places in the world with ideal year round temperatures between 40 and 60 degrees fahrenheit that facilitate a balanced mix of lifting while also skateboarding. Indoor skateparks are likewise either costly or crowded.

Outdoor skateboarding was halted during this time and several circuit exercises based on plyometrics, running, and power and olympic lifting are used to maintain endurance and build muscle volume (*Table 1*). The purpose was to improve skateboard ability by healing overworked muscle groups. Increasing gluteus maximus, upper thigh, and chest volume were the primary focus. Another target included improving ancillary posture groups such as the upper back and shoulders to counteract the somewhat slouched superficial appearance of the upper anterior torso

after mostly skateboarding for seven months. Freestyle skateboarding is also included as it requires a smaller space allotted by basements and retains coordination with a board.

Day/Group/ Order	Exercise	Description	Time
i/0/0	Warmup		10
i/0/1	Run	Fast one mile	10
i/0/2	Skate	*Current*: Old school kickflip, heelflip, inward heel, varial. *Retired*: primo, caspar stall, primo flip, finger flip, truck stand, no hand truck stand.	30
<exercise>	<exercise>		45
<exercise>	<exercise>		15
i/0/5	Run	One mile run	10
i/0/6	Stretch	Lay on back, stretch leg with band	15
i/1/0	Walk		25
i/2/0	Walk		25
i/3/0	Walk		25
i/3/1	Stretch		30
		Exercise List for <exercise>	
0/0/3	Legs 0	(Circuit) back squat 4X6 -> jump squats -> stepups -> box jumps. Good morning.	25
0/0/4	Core 0	Laying bicycle, infinity, mid leg raise motions, upright leg crunch (reverse), leg swings, oblique	25

		crunch.	
1/0/3	Arms 0	(Circuit) incline press -> rotating shoulder press -> tricep extension. Dumbbell flys -> anterior raises -> lateral raises.	30
1/0/4	Split Stretch	(Isometric flexes) hamstring bend -> one leg down dog -> cobra -> childs -> 1 leg cat -> lunge -> forward splits.	25
2/0/3	Legs 1	Power clean 3X10, push press, (circuit) pull ups -> bent rows.	30
2/0/4	Core 1	Planks, plank superman or bird dog, situps.	
3/0/3	Legs 2	(Circuit) back squat 3X10 -> bulgarian squat -> box jumps. Single leg dumbbell deadlift.	
3/0/4	Core 0		
4/0/3	Medial Legs	Walk, long warmup, throws, medial toe circle tennis ball, caspar stall, pullup hang, soleus lunge side lunge, horse stance, lotus.	
4/0/4	Arms 1	(Circuit) close grip press -> chest dips. Hang crunch, band shoulder rotations, curls.	
5/0/3	Core 1		
6/0/3	Arms 2	Snatch, overhead squat, upright rows.	
i/1/0	Juggling		5

Table 1: The first three and last exercise is completed as often as possible. The <exercise> value is filled with two exercises from the Exercise List based on the first value in the first column. The second value in the first column denotes the exercises completed throughout the rest of the day.

Indoor Skateboard Exercises

While lifting weights to fix muscle imbalances from primarily skateboarding, it proved difficult to accurately calculate the necessary kilocaloric (kcal) intake for resting metabolic rate

(RMR) and resulted in weight gain. Scales are misleading in this regard since an excess of 25 servings of water per day were used for hydration while using between 800 and 1200 kcal per day in training circuits. This justifies the less intense third phase where between 250 and 500 kcal were used every day learning new ollie tricks indoors on carpet. Since jumping is more difficult with more weight, attention was paid to removing excess weight from the discarded muscular repair materials circulating the body and resting.

From performing body weight exercises, it was discovered that moving the gastrocnemius concentrically with hip extension and knee flexion eliminated the remaining medial calf pain. The muscles posterior to the ankle and bottom of the foot became sore and were similarly alleviated. Dragging the sore leg's toes across the board with mostly imobile hip flexion and sudden knee extension identical to the ollie flip tricks also stretched the anterior lower leg muscles. Pistol squats, downdog inchworm, horse stance, and wall sit variations are used to mimic downhill and distance longboarding. A less intense weight lifting exercise is weighted side lunges depending on the foot position but does not increase ankle flexion or extension mobility as extremely as longboarding. Handstand pushups, pullups, and negative pushups are designed to increase shoulder and chest flexibility while bearing weight.

Day/Group/ Order	Exercise	Description	Time
0/0/0	Warmup 0	Rugby Throws	5
2/0/0	Warmup 1	Juggling	5
4/0/0	Warmup 2	Overhead throws	5
i/0/1	Skate Warmup	Revert, dog walk, one leg heel stretch, 180 board flip	10

i/0/2	Skate	OS flip, RL kickflip and heelflip, R manual kickflip and heelflip pose	45

0/0/3	Body Weight 0	Wall handstand pushup, negative pushup with dumbbell (circuit)	30
2/0/3	Body Weight 1	Pistol squat, downward dog/inch worm flexing heel into ground, side wall sits or longboard cruise (circuit)	30
4/0/3	Body Weight 2	Wall sit or downhill longboarding, pullups, horse stance (circuit)	30

i/0/4	Cool Down	Breathe while laying on the ground and isometrically flexing various muscles	10

i/1/0	Walk		25
i/2/0	Walk		25
i/2/1	Stretch	Laying on side of the bed: upper quad, back of knee, raised hamstring, knee to chest with a band	30
i/3/0	Walk		25
i/3/1	Stretch		30

Table 2: Wamup 0-2 are to reduce soreness from the Body Weight 0-2 exercises and is rationalized by not being the explicit muscles used. The other exercises are performed as often as possible as denoted with 'i'.

More attention is made to growing specific muscles with three phases of skateboard training as instigated through climate. Combining longboarding, skateboarding, and several low intensity walks with full stretches dramatically increased flexibility and mental stamina from 2024 until 2025. Although there were several shorter breaks from skateboarding during the summer (*Chapter 12 - Figure 27*), there was more unrecorded pain from muscular overuse observed in text notes at the end of 2024 and 2025. Chronically sore lower vastus lateralis

followed by the medial soleus one year later were dependent on the cold winter climate. The lifting body type in winter is very different from someone who skateboards every day. The most important discovery was concentrically moving the gastrocnemius to apply more pressure to the smaller ankle and foot muscles and justifying the ollie flip movements that elongate the anterior lower leg muscles.

Discussion

Knee pain emerged from imbalances between ankle supination and pronation, thigh abduction and adduction, and gluteal muscles during athletics and skateboarding was used to identify and correct the issues. Extensive data collection about activity scheduling, muscular soreness, and food intake led to increased understanding of hydration strategies. Irrespective of ankle sprains, audible distal medial ankle noise continued after ball kicking cessation and was ruled out as the cause. The right gastrocnemius and soleus was consistently sore after extended periods of walking, running, and lifting while dehydrated and is posture related. Weight during back squats shift from center to right in the descending leg flexion phase and from right to left in the extension phase indicating weaker left posterior abductors and right anterior adductors. Right lower back pain emerged from high school athletics and was reduced from longboarding but reappeared while mostly walking. Head and neck pain originates from both exercising or mental strain combined with a lack of nutrient instigated dehydration, mental stress, and muscular imbalance. There are several unrelated activities that assist skateboarding.

The intention of collecting data for 100 more days in Chapter 12 was to find a statistical relationship between activities, nutrition, and observed pain to mathematically predict debilitating soreness to improve skateboard ability. Hydration was cemented as important to avoid muscle soreness leading to training disrupting headaches. Hydration difficulties emerged after training since water is eventually unappetizing and diet strategies from school athletics to induce thirst were utilized. Hydration and the efficient extraction of minerals such as salt through intaking enough potassium and fiber is difficult to complete while on a schedule. Overcorrecting inadequate dietary concerns with more plants can also lead to variable blood sugar or low salt

and result in head pain. Nutrition was a focus by ingesting enough potassium to remove excess salt and water from post-workout recovery meals without raising blood sugar from excessive carbohydrates.

Right anterior and posterior ankle snapping emerged while skateboarding and right footed ball kicking. To test if the right ankle was damaged from previous ankle sprains, left footed ball kicks were practiced and produced similar left anterior ankle snapping while the right anterior noise ceased. Right posterior medial ankle noise continued indicating a lack of muscular balance. Since right soleus muscle soreness occurred prior to a minor ankle injury, the right medial snapping ankle was from poor walking and running form. Strengthening the ankle muscles was performed with Casper stalls and rolling the ankle into the skateboard during manuals. Medial lower leg strength also increased while practicing downhill longboarding and from specific lifting exercises. Lower right calf pain was mostly eliminated from shortening the upper calf and flexing isometrically followed by lengthening. The concept is important to avoid redundantly lengthening the lower hamstrings and straining the upper back and neck or hip muscles with bends or sitting stretches. The resulting sore ankle and foot muscles were also stretched using this method and support strengthening the entire body.

Right lower back pain unfolded while learning back squats in 2009-2013. Left gluteus medius strength and flexibility improvements partially reduced the issue through rest and physical therapy in 2009 and was mostly rectified with regular stanced distance longboard pushing and lighter weight front squats from 2014-2016. Gluteus and hamstring to quadricep muscle imbalance was addressed from November 2024 to February 2025 with back squats and freeweight leg exercises. Squat form improved by rolling the hips and gluteus posteriorly and leaning slightly forward to limit knee snapping. Weight imbalance was observed for back squats

near exhaustion overrelying on the right gluteus medius in the descent and the left quadriceps abductors and right hamstring adductor in the ascent. Left lower back pain was observed while squatting with a mirrored technique and supports the stronger left anterior to right posterior leg strength theory for one of the origins of right lateral knee pain. Squat form was modified to limit this effect by using less weight, narrowing the grip, and pulling the bar into the trapezius to force forward lean and gluteus use.

More evidence for greater left anterior and right posterior leg strength mismatch includes posterior lower leg pain after skateboarding, lifting the left leg off the board during switch ollie tricks, rotating the right leg laterally while running, and medically documented lower right back multifidus pain while exercising from right limbed throwing, kicking, and swinging. The left hip also has greater anterior vascular definition. Longboard pushing and heelslides from 2014-2016 contributed to abductor and adductor imbalance that was exaggerated while working a desk job and mostly walking for exercise from 2017-2021. Right knee pain emerged in early 2022 while working a physically active night shift job. Knee pain became more apparent while increasing jogging distance from one to 15 miles during 2022 but was minimized while not running and relearning regular stance downhill longboarding from 2023-2024. The lack of right adductor balance caused irregular stride and accelerated the knee issue by rotating the femur and hyperextending the leg. Rear leg adductor strength was further exposed while learning regular stance ollies in 2024 and accentuated the existing right lower vastus lateralis pain.

Similar left knee snapping was revealed while learning switch ollie tricks and observing the rear left leg lifting off the board's tail lends extensive evidence to stronger left anterior muscles. Stretching the lateral leg muscles was the initial method of reducing inflammation and colloquial terms such as DBS led to further concentration on reducing the anteriorly rotated

pelvis that emerged while sitting for long periods with low exercise. Greater right anterior and left posterior leg muscles accumulated from pushing the longboard switch from 2024-2025. Right anterior adductor strength increased in 2025 from pulling the board perpendicular to the road during regular stance downhill longboard toe slides and runs conversely to the overdeveloped right posterior adductor from sweeping the leg forward in the first half of the heel slide. Muscular use is mimicked with freestyle skateboard tricks such as switch stanced Casper stalls and 5050 truck stands or traditional weight lifting exercises like lateral lunges and various angled wall sits.

Pain around the right skull's frontal bone originates from the posterior neck and is a result of either muscular exhaustion from exercise or mental strain while sitting extensively. Although nutrition was not thoroughly addressed, imbalance between nutrients leads to head pain from low hydration decreasing muscular elasticity or vitamin deficiencies disrupting efficient function resulting in the accumulation of smooth or skeletal muscle repair molecules. Skateboarding rolls the shoulders anteriorly and causes different muscular development between the chest and upper back leading to head pain. The effect is converse to back squats or pull ups that roll the shoulders posteriorly also causing head pain. Dehydration from inadequate post-exertion macro or micro nutrient intake is the primary cause and stems from uncontrollable secondary issues such as lack of time to prepare meals at rigid scheduling that makes extensive exercise monetarily unaffordable or too mentally exhausting for long-term enjoyment. There are several low movement activities as determined by demographic that assist in providing a mental reprieve from athletics while retaining responsiveness to adequately nourish and stretch. The participant's hobbies that were especially applicable to this skateboarding self-experiment included enjoying general exercise, software development to learn mathematical and science fundamentals for

accurately identifying bodily failures, and guitar to increase wrist strength and coordination. Juggling is useful for neck pain reduction with arm coordination, rotator cuff movement, and casual hamstring bends while collecting dropped objects.

Excessive documentation while skateboarding had several disadvantages such as increasing the amount of time obsessing about irrelevant details and decreasing time to relax. Using computer science fundamentals to perform data analysis provided a hiatus from skateboarding but is markedly different from learning microbiology for nutrition, neurobiology, or the latin terms for muscular anatomy and especially different from health care provider education. Abstracted mathematical concepts using C syntax are more closely related to operational efficiency research and revealed applied mathematical methods to understanding the musculoskeletal, nervous, and digestive systems. Managing the body's systems are programmed into general education but the ability to explicitly articulate are scaled after K-12. The implications for automated exercise forecasting has the potential for disastrously long lasting bodily injuries when applied to consuming media where behavior modifications are inserted. Elite Olympic style psychological motivation is annoying and can lead to mental instability. Automated behavior modifications based on ephemeral social science information systems has the potential for undesired effects.

Conclusions

Right knee pain is from hyperextending the leg while moving due to several muscular imbalances starting with right lower back pain from heavy back squats and continuing with longboarding induced quadricep and hamstring mismatch which were illuminated while primarily walking for exercise. Since right lower back pain emerged near the end of 2025 after pushing switch, it would make sense to resume regular stanced longboard pushing to strengthen the left gluteus medius. Knee pain moved to calf pain after practicing the kickflip and using downhill longboarding in 2025 to work different sections of the same muscle groups. Alternating between regular and switch tail and heel slides would retain balanced muscular development at all lengths to continue learning ollie tricks. Shortening the calves, posterior ankles, and feet muscles during downhill and through specific concentric stretches reduced anterior knee, posterior calf, and foot pain. Combining the subjective pain scale with objective measurements through non-parametric statistics resulted in low confidence and strategies to reduce inflammation requires individual awareness. Specific anatomy terms were learned with reasoning behind training regiments from athletics. Fixing one body area is difficult because another body part will be over or under used resulting in soreness.

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